

TASK 3 REPORT

JWT Authentication and Protected Routes

Project Title

Student Management System using MERN Stack with JWT Authentication

Objective

The objective of this task is to implement user authentication and authorization using JSON Web Tokens (JWT). This ensures that only authenticated users can access protected resources within the application.

Technologies Used

- MongoDB
- Express.js
- React.js
- Node.js
- JWT (JSON Web Token)
- bcryptjs
- Postman

Features Implemented

1. User Registration (Signup)

- Created a signup API endpoint.
- Users can register using name, email, and password.
- Passwords are encrypted using bcryptjs before storing in MongoDB.
- Duplicate email validation is implemented.

2. User Login

- Created a login API endpoint.
- Users can login using registered email and password.
- Password verification is performed using bcryptjs.
- JWT token is generated after successful authentication.

3. JWT Authentication

- JWT token is generated using a secret key.
- Token contains user identification information.
- Token expiration time is set to 1 hour.

4. Middleware Protection

- Created a custom JWT verification middleware.
- Middleware checks Authorization header.
- Invalid or missing tokens return an Access Denied message.
- Valid tokens allow access to protected routes.

5. Protected Dashboard Route

- Created a protected dashboard endpoint.
- Only authenticated users can access this route.
- Unauthorized users receive an Access Denied response.

6. Frontend Integration

- Created Signup page.
- Created Login page.
- Created Dashboard page.
- Stored JWT token in browser Local Storage.
- Implemented ProtectedRoute component.
- Users are redirected to Login page if token is missing.

API Endpoints

POST /signup

Request Body:

```
{
  "name": "User",
  "email": "user@example.com",
  "password": "123456"
}
```

Response:

```
{
  "message": "User Registered"
}
```

POST /login

Request Body:

```
{
  "email": "user@example.com",
  "password": "123456"
}
```

Response:

```
{
  "token": "JWT_TOKEN"
}
```

GET /dashboard

Response:

```
{  
  "message": "Welcome to Protected Dashboard"  
}
```

Testing Performed

- Signup API tested using Postman.
- Login API tested using Postman.
- JWT token generation verified.
- Protected route tested with valid token.
- Protected route tested without token.
- React frontend authentication verified.
- Local Storage token storage verified.

Results

The JWT Authentication system was successfully implemented. Users can register, login securely, receive JWT tokens, and access protected routes only after successful authentication.

Conclusion

Task 3 was completed successfully by implementing JWT-based authentication and protected routes in the Student Management System. The application now provides secure user access control and authorization functionality using the MERN stack.